Project Development Phase

Date	**06-03-2026**
Team ID	**Srasti Patel**
Project Name	**Heart Disease Analysis**

1. Introduction to Tableau Visualization

Tableau was used to transform the processed heart disease dataset into meaningful visual insights. The cleaned dataset was imported into **Tableau Desktop Public Edition**, where multiple worksheets were created to analyze relationships between different health indicators and heart disease occurrence.

The visualizations help identify patterns related to **age, gender, diabetes, smoking habits, physical activity, BMI levels, and general health conditions**. These visual insights allow healthcare professionals and analysts to better understand the key factors contributing to heart disease.

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2. Tableau Worksheets (Visualizations)

2.1 Worksheet 1 – Gender vs Heart Disease

Visualization Type: Stacked Bar Chart

Business Question:
Is heart disease more prevalent among males or females?

Description:
This visualization compares the count of heart disease cases between male and female individuals. The stacked bars show the distribution of patients with and without heart disease based on gender.

Insight:
The visualization indicates that heart disease prevalence is slightly higher among males compared to females.

Screenshot:

2.2 Worksheet 2 – Age vs Heart Disease

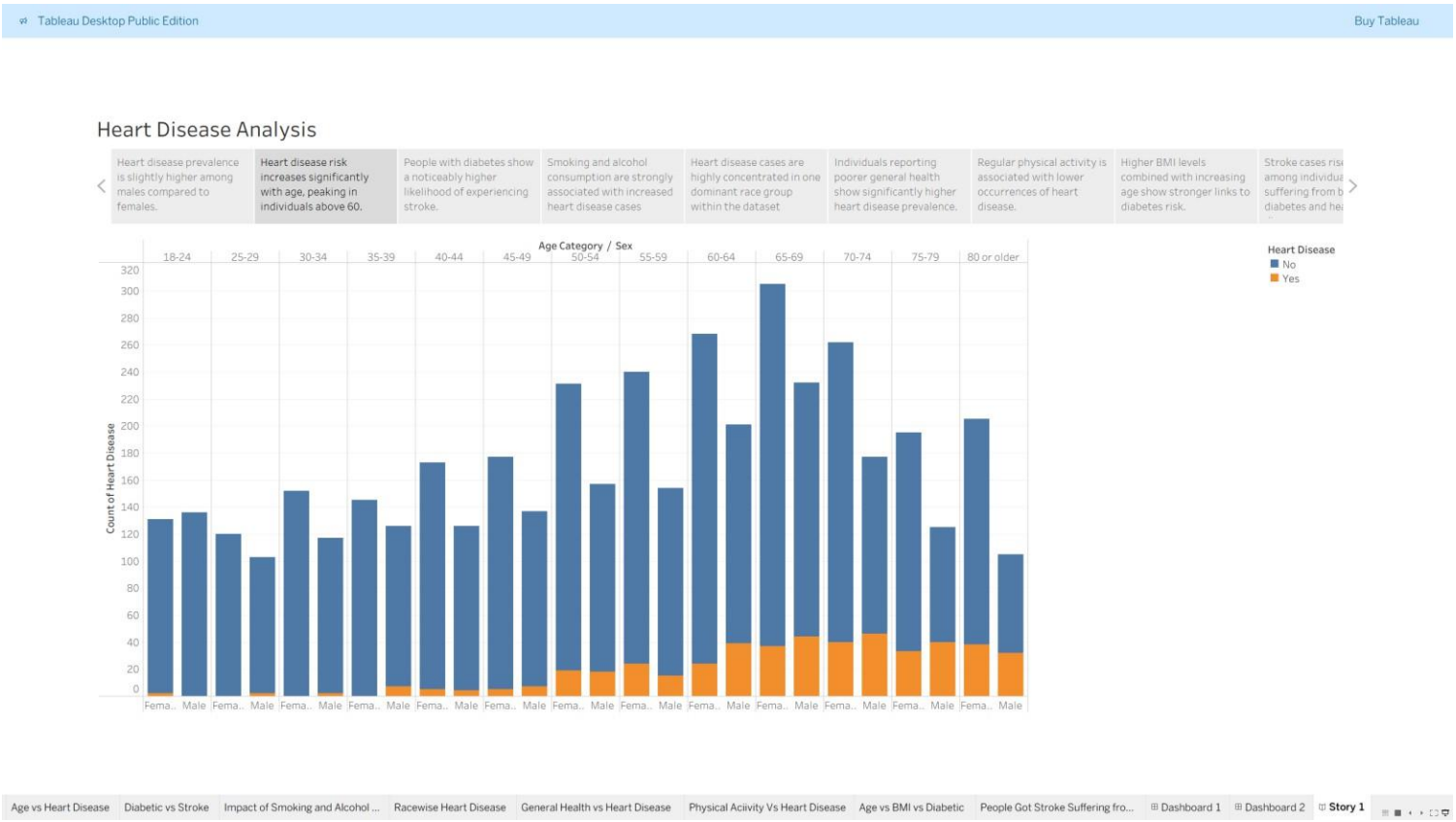
Visualization Type: Bar Chart

Business Question:
How does heart disease risk vary across different age groups?

Description:
This chart analyzes heart disease cases across multiple age categories. It highlights how the number of heart disease cases changes as age increases.

Insight:
Heart disease risk increases significantly with age, with the highest number of cases observed in individuals above 60 years.

Screenshot:



2.3 Worksheet 3 – Diabetes vs Stroke

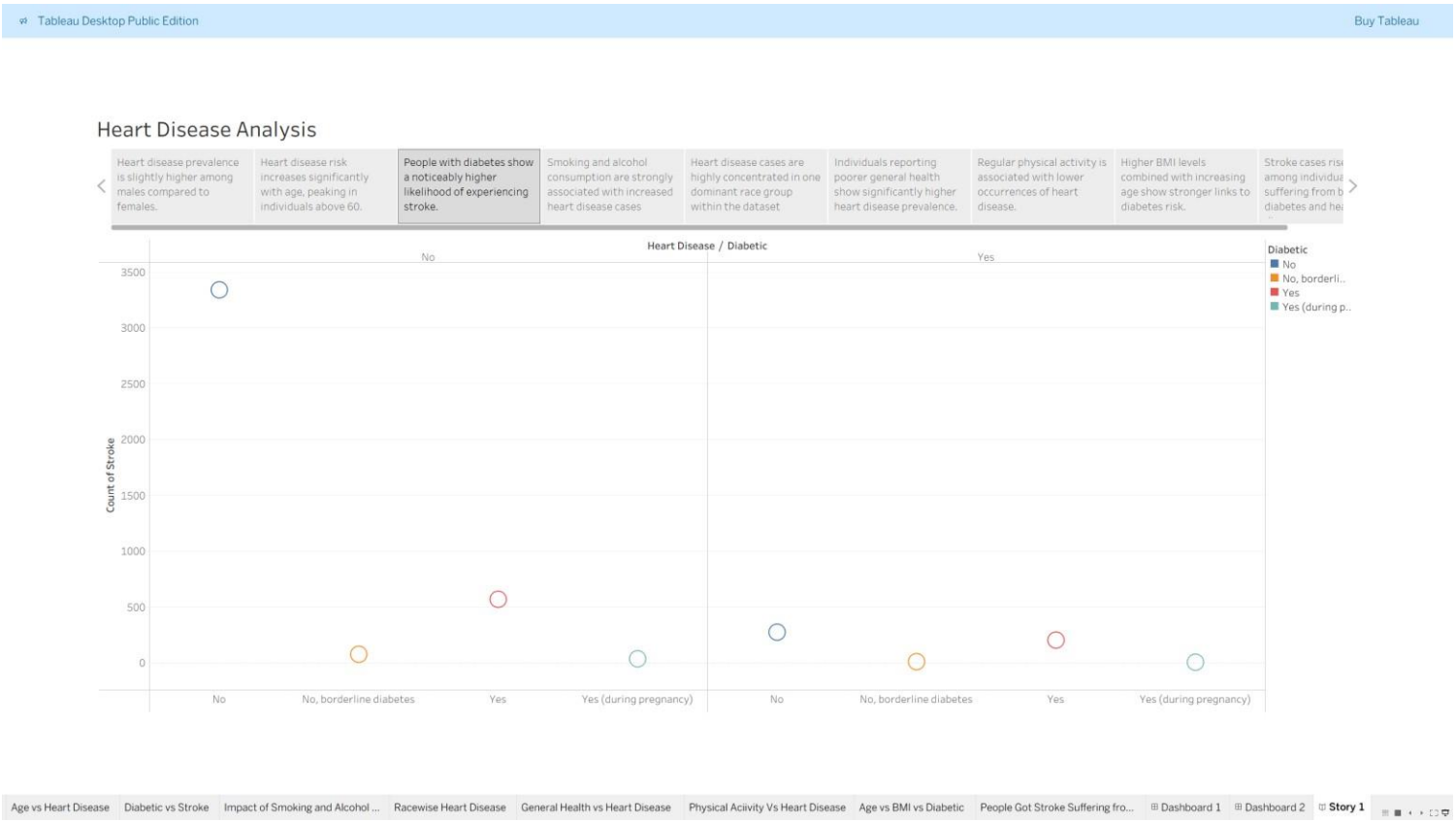
Visualization Type: Bubble / Scatter Plot

Business Question:
Are individuals with diabetes more likely to experience strokes and heart disease?

Description:
This visualization compares the count of stroke cases among individuals with different diabetes conditions. It shows the relationship between diabetes and stroke occurrence.

Insight:
People diagnosed with diabetes show a noticeably higher association with stroke and heart disease.

Screenshot:



2.4 Worksheet 4 – Smoking and Alcohol Impact

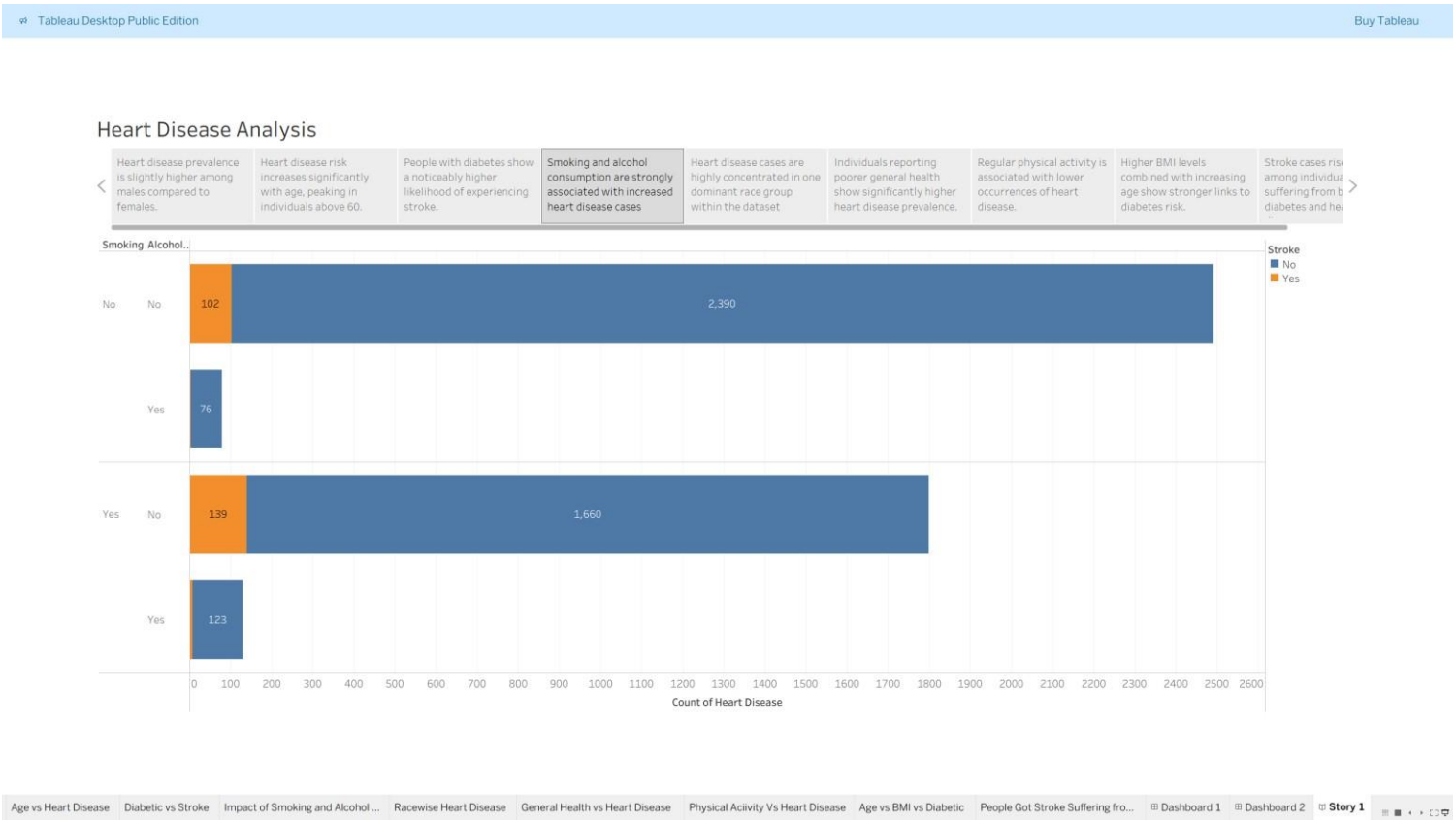
Visualization Type: Horizontal Stacked Bar Chart

Business Question:
Does smoking and alcohol consumption increase the risk of heart disease?

Description:
This visualization analyzes heart disease cases based on smoking and alcohol consumption habits.

Insight:
The chart shows that individuals who smoke or consume alcohol have a higher likelihood of developing heart disease compared to those who do not.

Screenshot:



2.5 Worksheet 5 – Race-wise Heart Disease Distribution

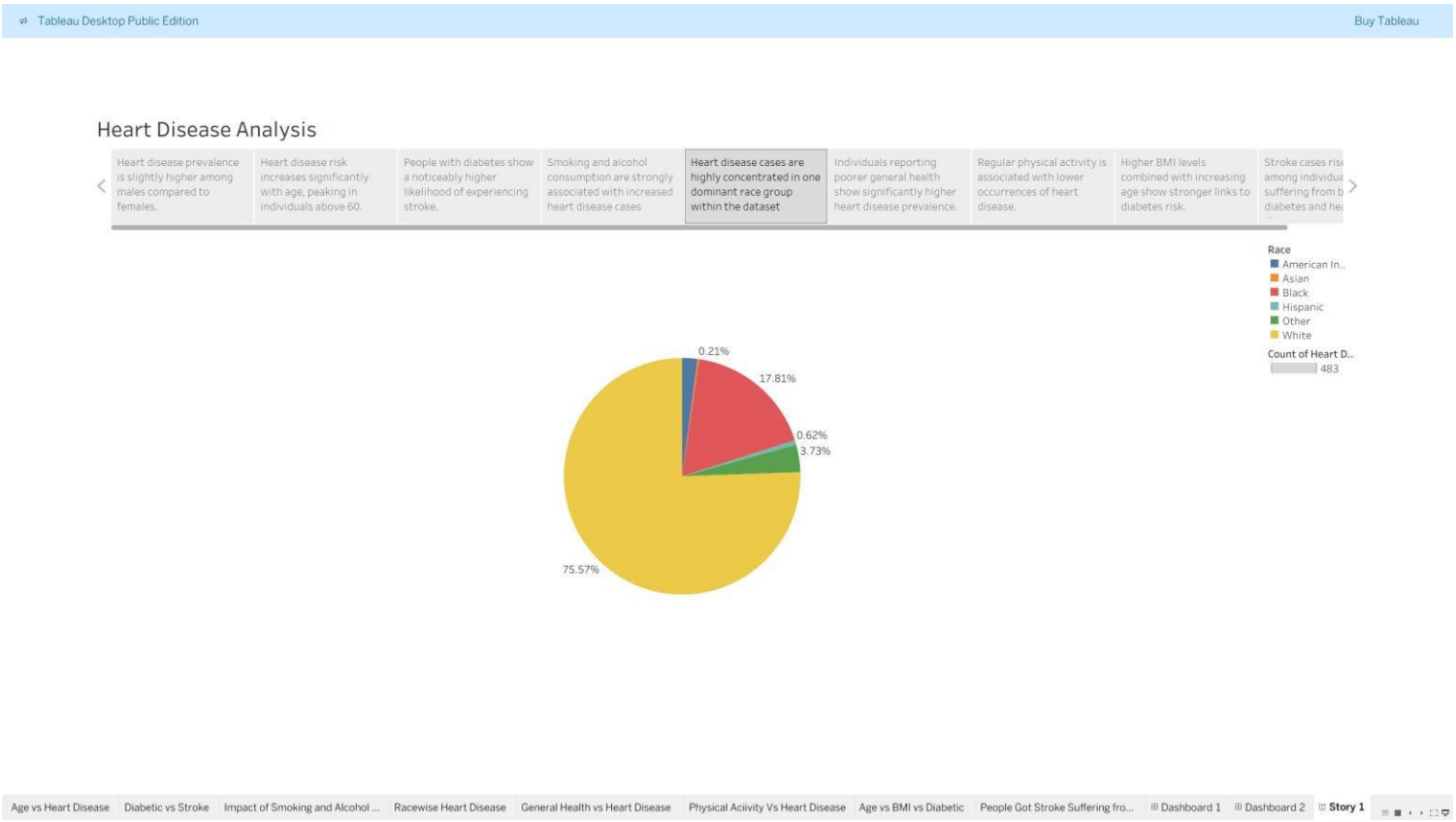
Visualization Type: Pie Chart

Business Question:
Which racial group has the highest percentage of heart disease cases?

Description:
This visualization displays the percentage distribution of heart disease cases among different racial groups.

Insight:
The analysis shows that a dominant racial group accounts for the majority of heart disease cases in the dataset.

Screenshot:



2.6 Worksheet 6 – General Health vs Heart Disease

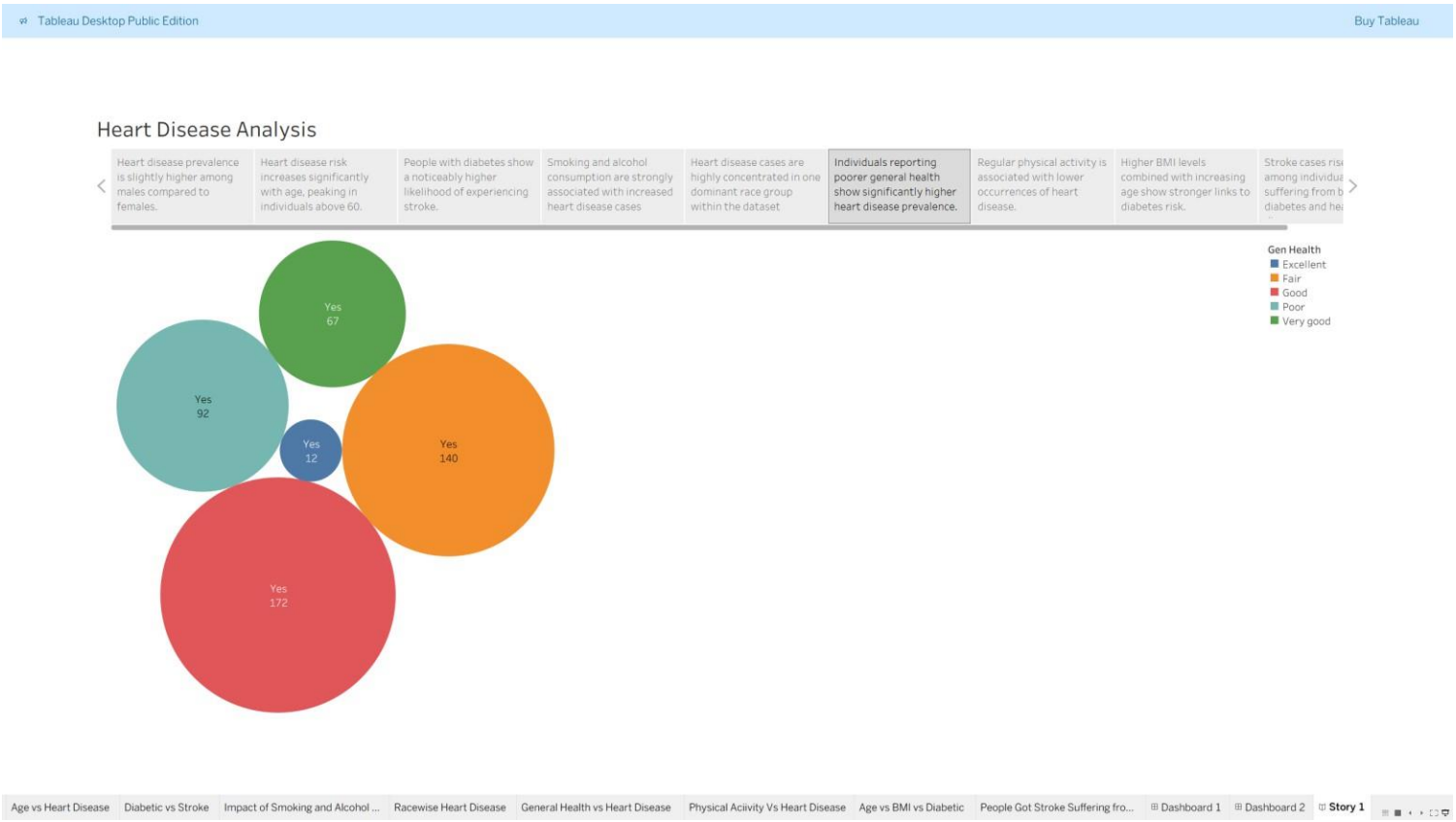
Visualization Type: Bubble Chart

Business Question:
How does general health status influence heart disease prevalence?

Description:
This visualization represents the relationship between self-reported general health conditions and heart disease cases.

Insight:
Individuals reporting poor or fair general health show significantly higher occurrences of heart disease.

Screenshot:



2.7 Worksheet 7 – Physical Activity vs Heart Disease

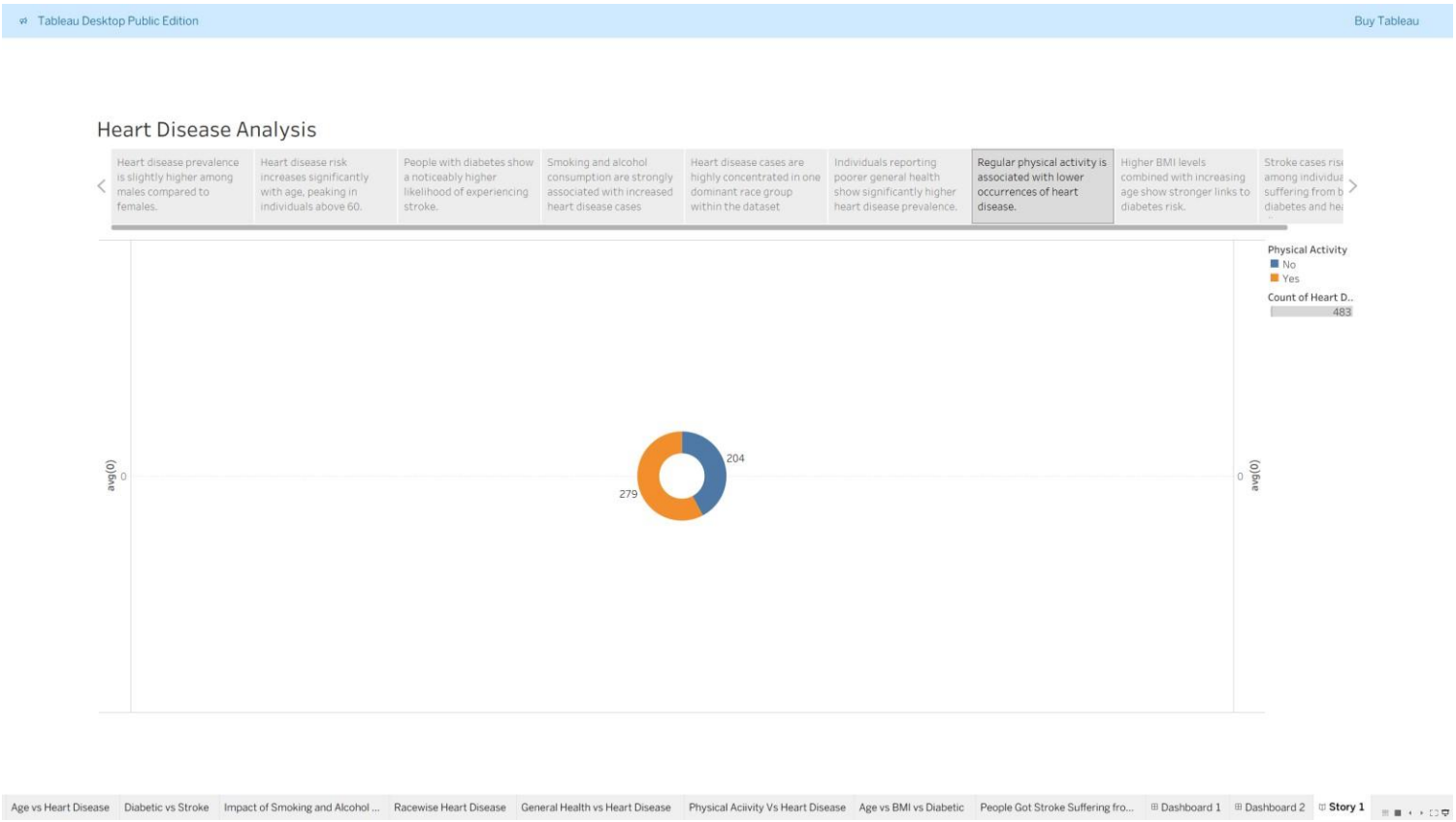
Visualization Type: Donut Chart

Business Question:
Does regular physical activity reduce the risk of heart disease?

Description:
This visualization compares heart disease occurrences among individuals who perform regular physical activity versus those who do not.

Insight:
Individuals who engage in regular physical activity show a lower occurrence of heart disease compared to inactive individuals.

Screenshot:



2.8 Worksheet 8 – Age vs BMI Analysis

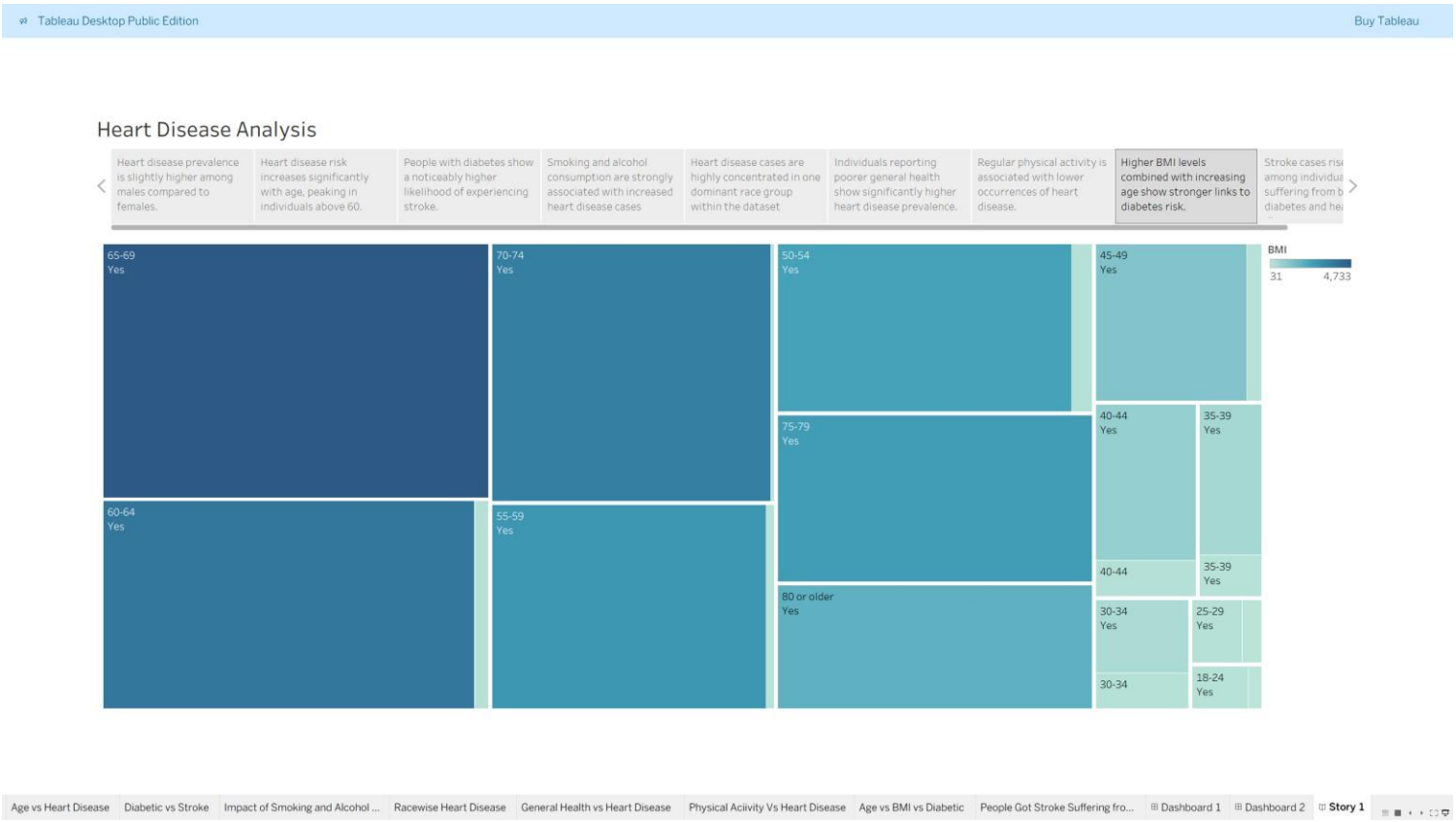
Visualization Type: Treemap

Business Question:
How do BMI levels vary across different age groups?

Description:
This treemap shows BMI distribution across multiple age categories, helping to identify age groups with higher BMI levels.

Insight:
Higher BMI values combined with increasing age are associated with higher heart disease risk.

Screenshot:



2.9 Worksheet 9 – BMI vs Diabetes vs Age

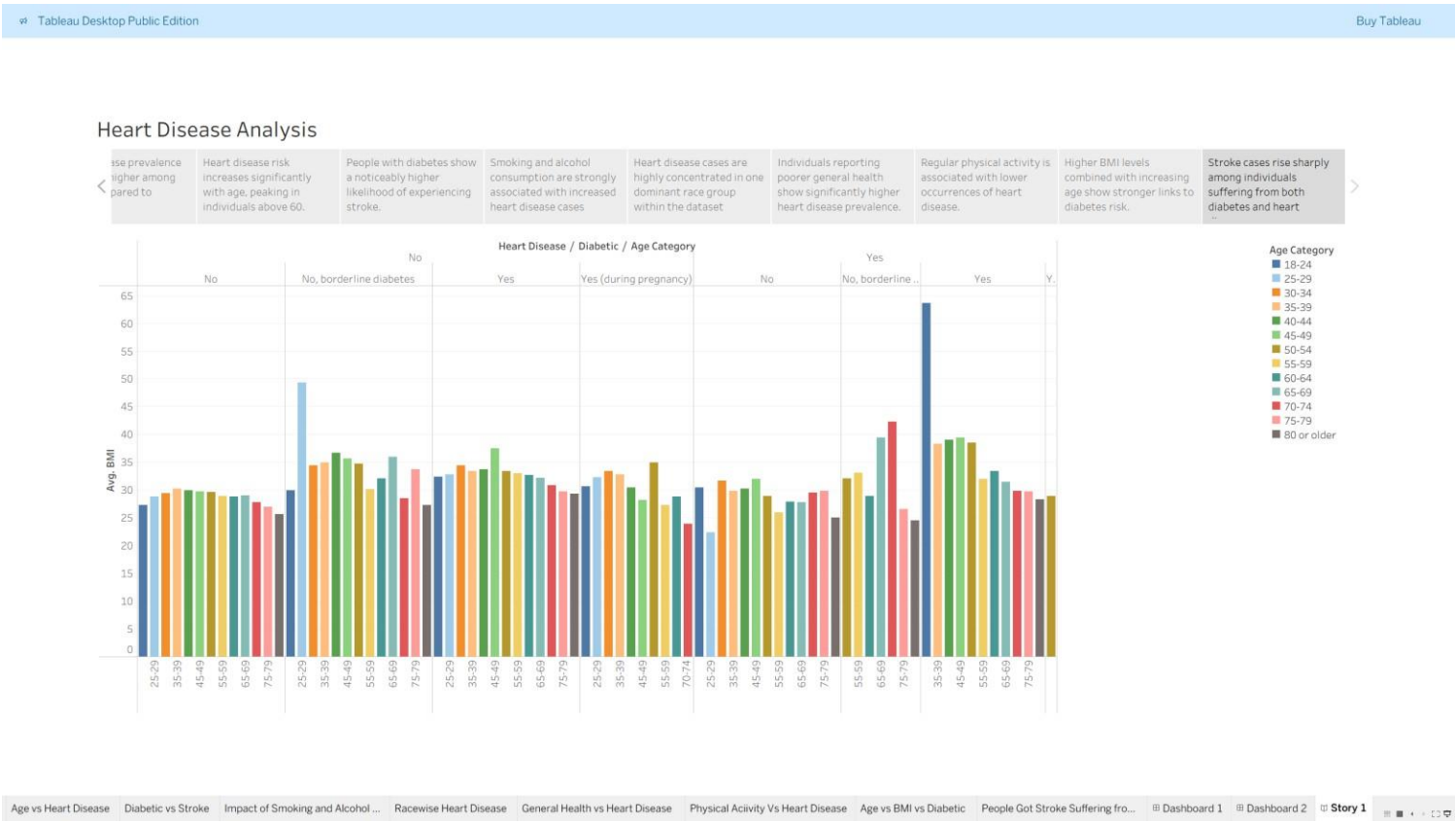
Visualization Type: Multi-category Bar Chart

Business Question:
How does BMI vary across age groups and diabetes conditions in relation to heart disease?

Description:
This visualization analyzes BMI levels across age groups and diabetic conditions, helping identify patterns related to heart disease risk.

Insight:
Higher BMI levels are more common among individuals with diabetes, especially in older age groups.

Screenshot:



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3. Tableau Dashboard

3.1 Dashboard Overview

After creating individual worksheets, they were integrated into an interactive **Tableau Dashboard** to provide a comprehensive view of heart disease patterns.

The dashboard allows users to explore multiple health indicators simultaneously and understand how different factors influence heart disease risk.

3.2 Dashboard Features

- Interactive visualization environment
- Multiple health indicators analyzed together
- Clear graphical representation of heart disease trends
- Easy comparison of different demographic and health factors

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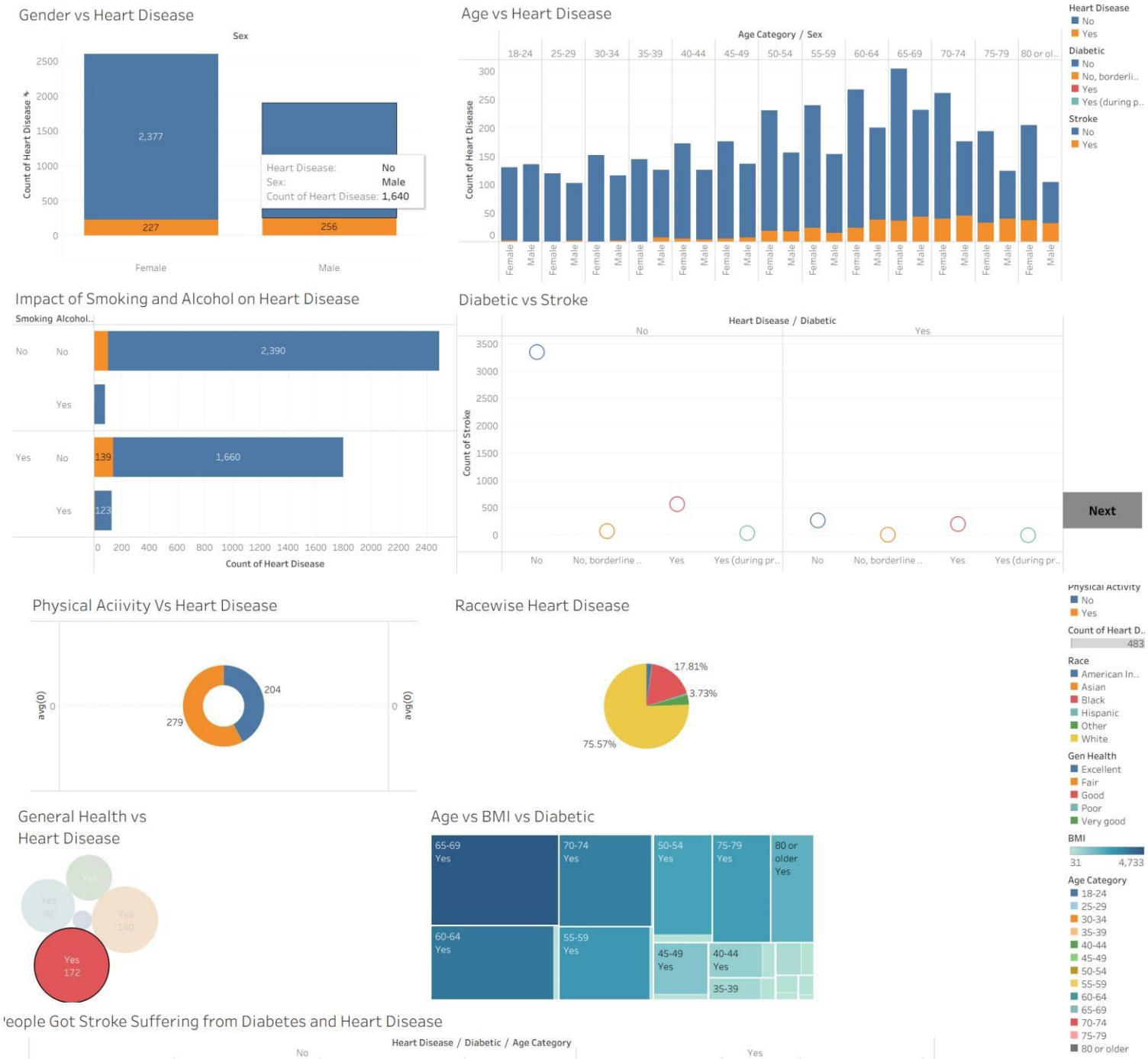
3.3 Dashboard Screenshot



4. Tableau Story

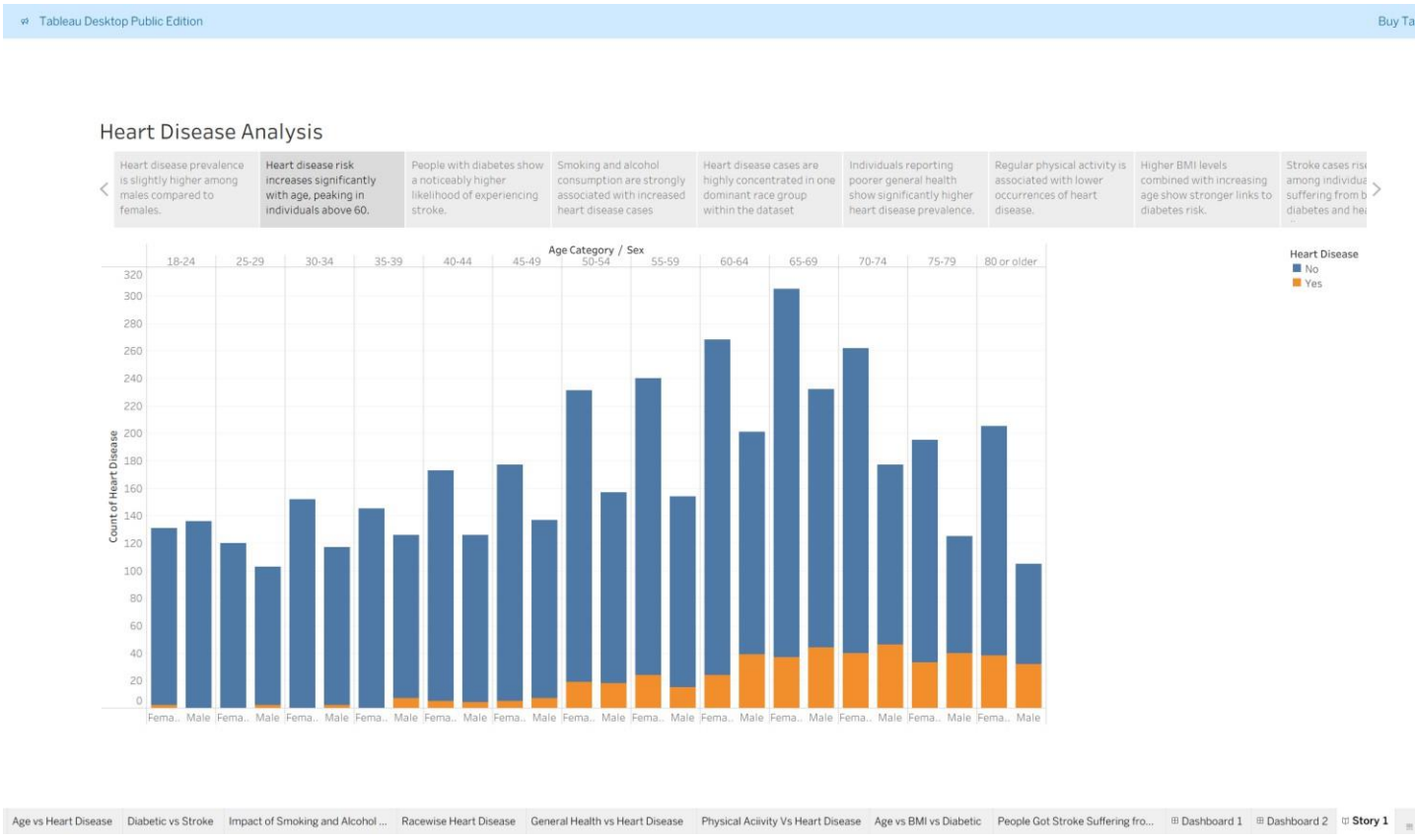
4.1 Story Overview

A **Tableau Story** was created to present the visual insights in a structured sequence. The story guides users through key analyses and findings related to heart disease risk factors.

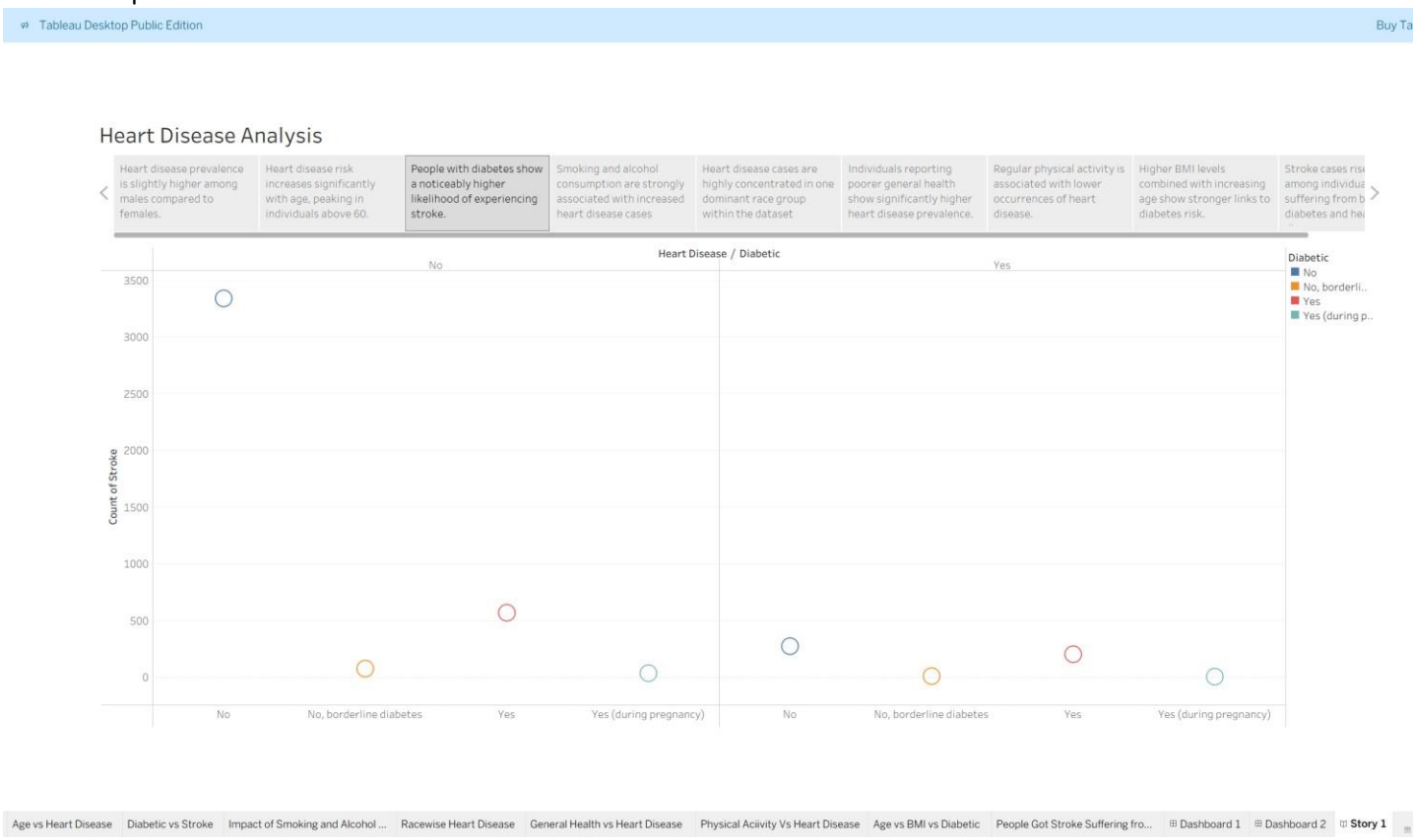
4.2 Story Points/4.3 Story Screenshot

1. Gender-based heart disease comparison

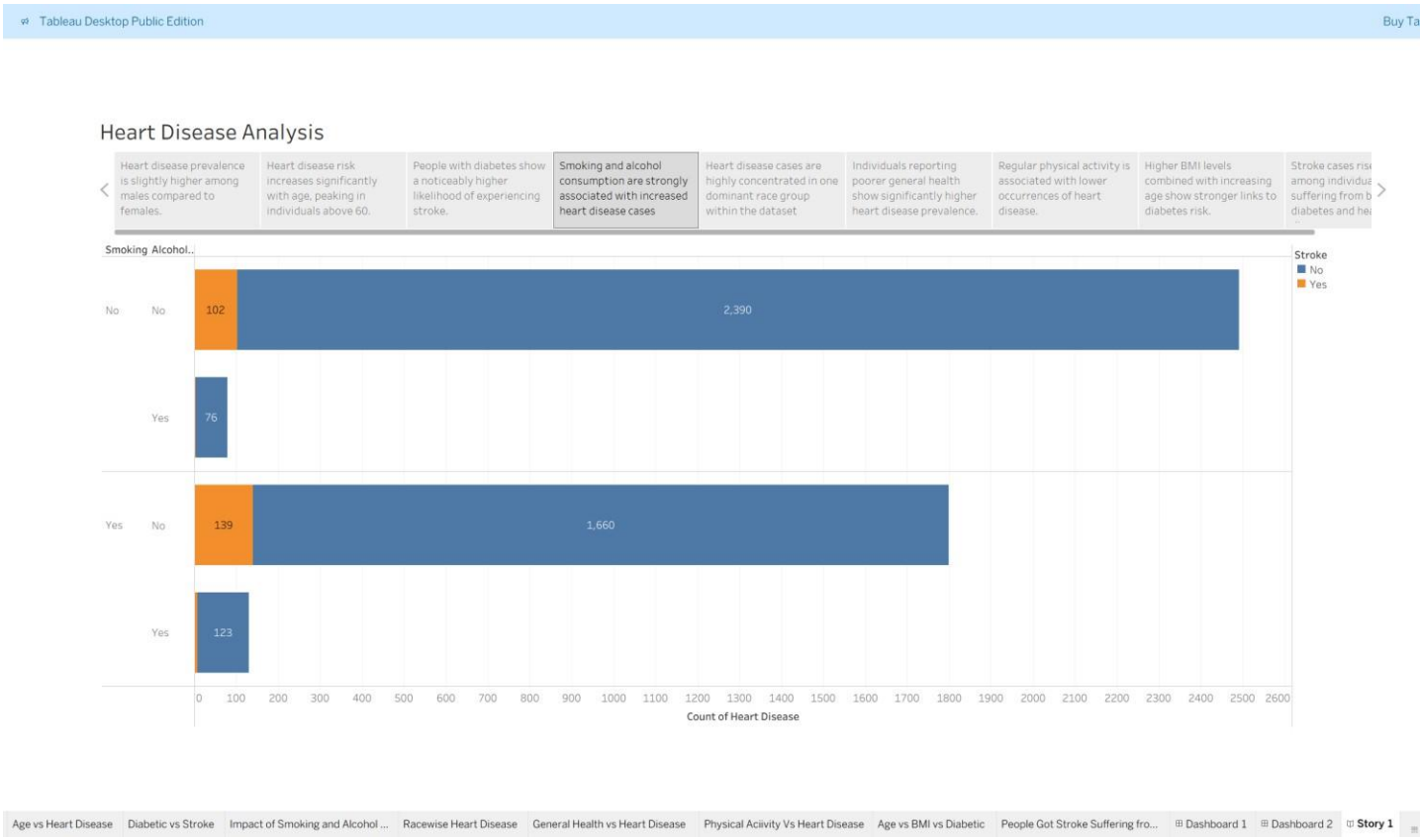
2. Age-related heart disease risk analysis



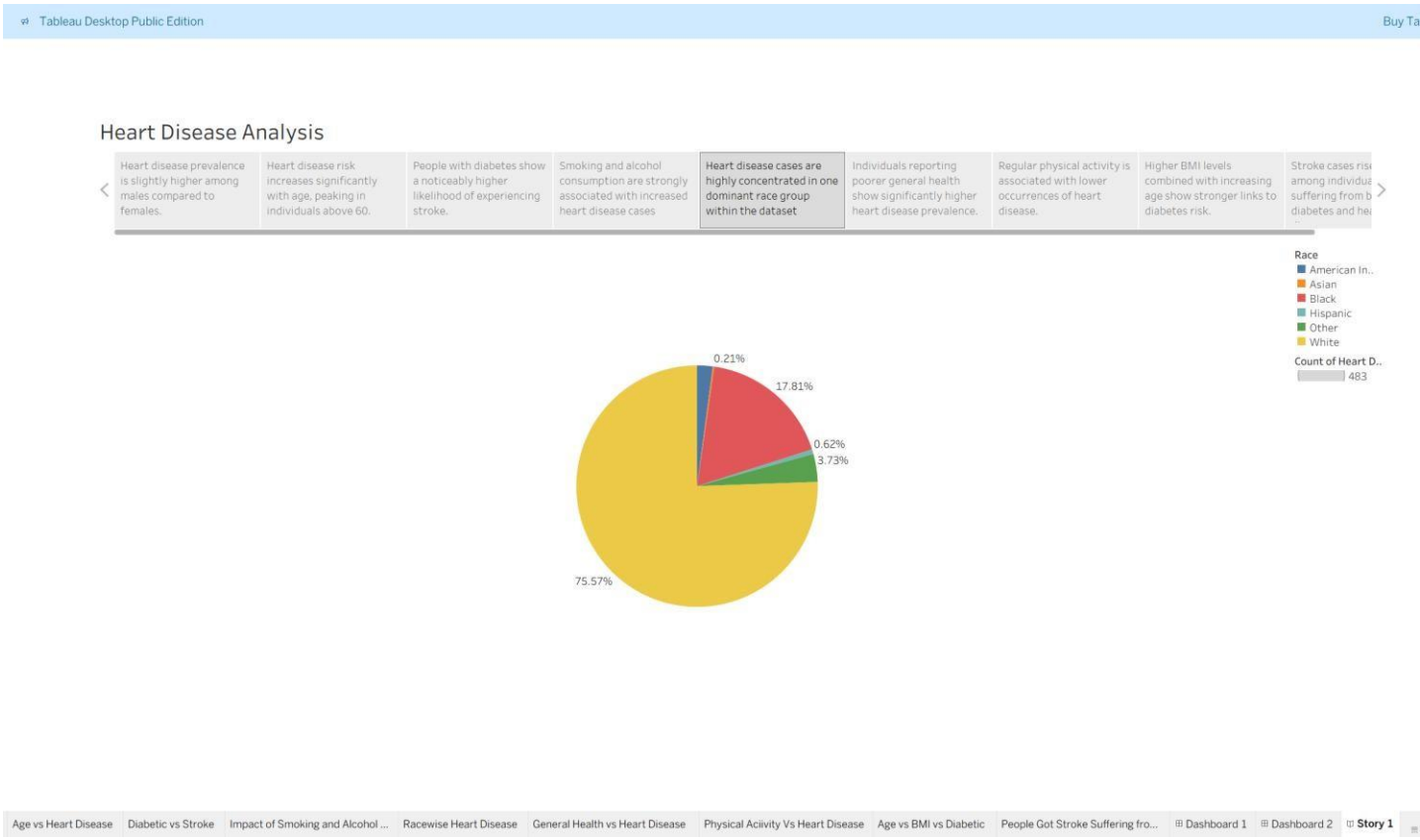
3. Relationship between diabetes and stroke



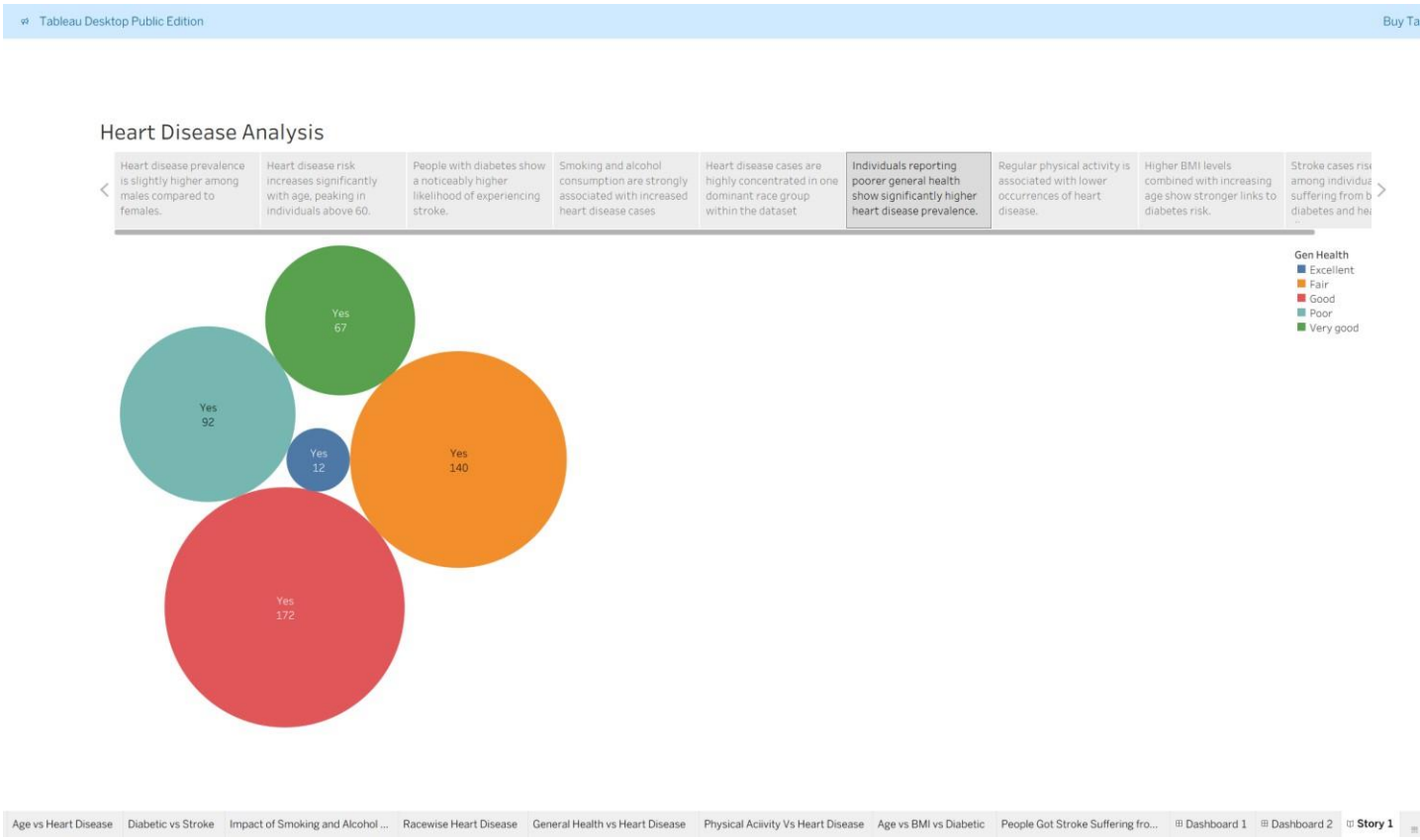
4. Impact of smoking and alcohol consumption



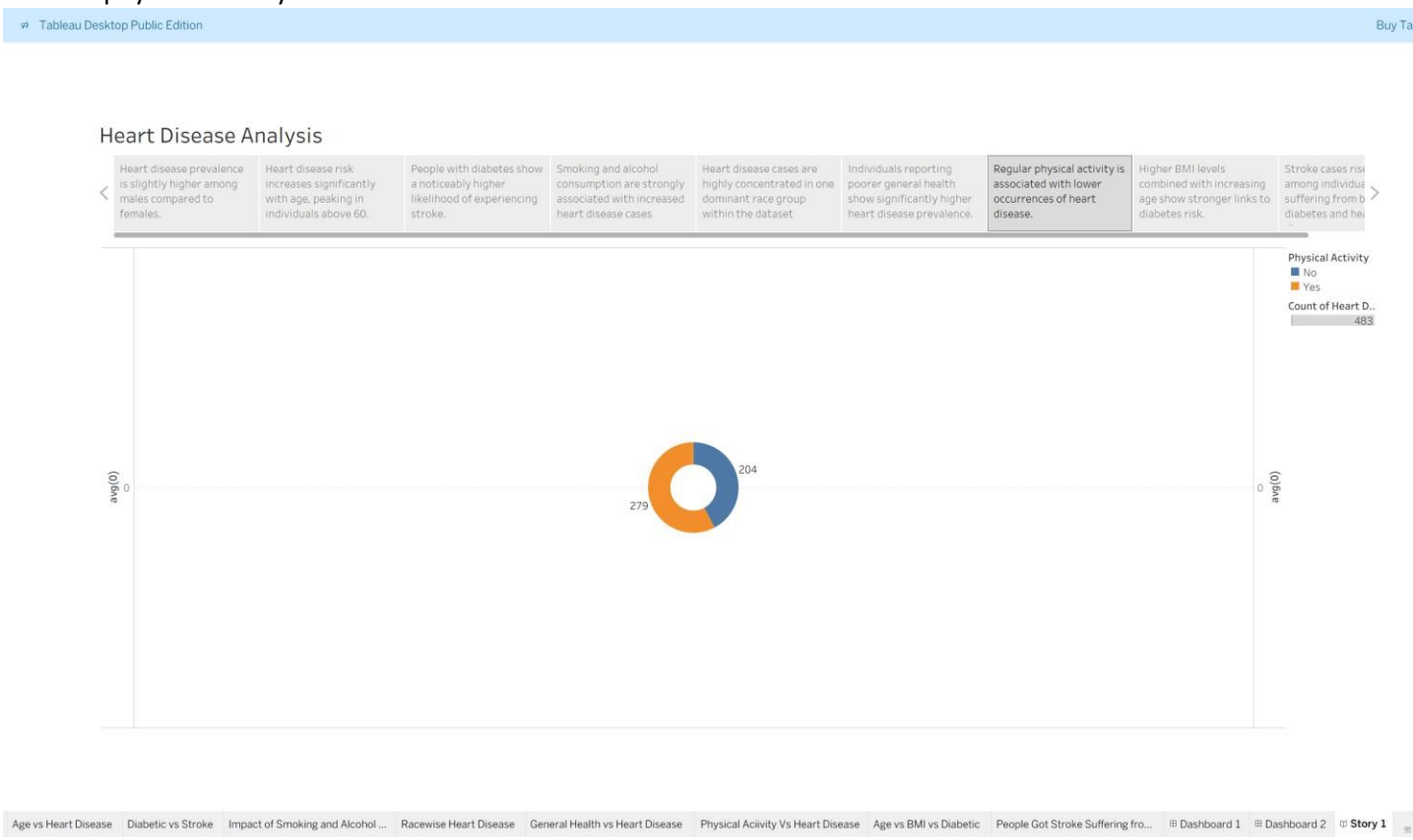
5. Race-wise distribution of heart disease



6. Influence of general health conditions



7. Effect of physical activity on heart disease



8. Impact of BMI and age on health risks

5. Tableau Public Deployment

5.1 Publishing Process

After completing the dashboard and story in Tableau Desktop, the project was published on **Tableau Public**. Publishing allows the dashboard to be accessed online and shared with others for interactive analysis.

5.2 Tableau Public Link

The final project dashboard is available through the following Tableau Public link:

Tableau Public Project Link:

https://public.tableau.com/views/HeartDiseaseAnalysis_17728814529840/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Story Link:

https://public.tableau.com/views/HeartDiseaseAnalysis_17728814529840/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

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6. Summary of Visualization Insights

The Tableau visualizations provide valuable insights into heart disease patterns and risk factors. The analysis highlights how age, lifestyle habits, health conditions, and demographic characteristics influence heart disease prevalence.

By presenting the data through interactive dashboards and stories, the project enables a clear understanding of complex healthcare data and supports better decision-making for healthcare analysis.